

SOLIDWORKS API (.NET, VBA, C#) — Reference (285/337): ISldWorks (part 5 of 8)

SOLIDWORKS API Documentation (converted from CHM)

ISldWorks (part 5 of 8) (Type)

SOLIDWORKS API Help

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IActivateDoc3 Method (ISldWorks)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IActivateDoc3 Method (ISldWorks)

Activates a document that has already been loaded. This file becomes the active document, and this method returns a pointer to that document object.

.NET Syntax

Visual Basic (Declaration)

```
Function IActivateDoc3(  
    ByVal Name As System.String, _  
    ByVal Silent As System.Boolean, _  
    ByRef Errors As System.Integer _  
) As ModelDoc2
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim Name As System.String  
Dim Silent As System.Boolean  
Dim Errors As System.Integer  
Dim value As ModelDoc2
```

```
value = instance.IActivateDoc3(Name, Silent, Errors)
```

C#

```
ModelDoc2 IActivateDoc3(  
    System.string Name,  
    System.bool Silent,  
    out System.int Errors  
)
```

C++/CLI

```
ModelDoc2^ IActivateDoc3(  
& System.String^ Name,  
& System.bool Silent,  
& [Out] System.int Errors  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Name of document to activate

Silent

True if dialogs and warning messages should be avoided; false if dialogs and warning messages should be displayed to the end-user

Errors

Status of the document activate operation as defined in [swActivateDocError_e](#); if no errors or warnings are encountered, this value is set to 0

Return Value

[Model document](#)

Visual Basic for Applications (VBA) Syntax

See [SldWorks.IActivateDoc3](#).

Example

[Access Assembly \(C++\)](#)

[Add Component and Mate \(C++\)](#)

Remarks

This file becomes the active document, and this method returns a pointer to that document object.

If you do not specify a file extension in the name parameter, the document activated is based on its name and ignores the file extension. This can cause problems if you have two different document types with the same name loaded; for example, 12345.sldprt and 12345.sldasm.

If you do not specify the filename extension in your call to this method, then you cannot be sure which document is activated. To avoid this problem, you can specify the file name extension in the name parameter or you can check the document type after it is activated using [IModelDoc2::GetType](#).

In some instances, a document requires a rebuild when it is activated. If this is the case and you have passed True for the silent argument, then the activated document is not rebuilt and swDocNeedsRebuildWarning is returned in the errors argument. You can then programmatically rebuild the document using the [IModelDoc2::EditRebuild3](#) method.

If the document needs a rebuild upon activation and you have passed false for the Silent argument, then a dialog is displayed to the users asking them if they want to rebuild the document.

If the user answers...	Then the activated document is...
No	Not rebuilt and swDocNeedsRebuildWarning is returned in the errors argument.
Yes	Rebuilt immediately and this warning is not returned.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::ActivateDoc2 Method](#)
[ISldWorks::ActiveDoc Property](#)
[ISldWorks::IActiveDoc2 Property](#)
[ISldWorks::CloseAllDocuments Method](#)
[ISldWorks::CloseDoc Method](#)
[ISldWorks::GetFirstDocument Method](#)
[ISldWorks::GetOpenDocument Method](#)
[ISldWorks::GetOpenDocumentByName Method](#)
[ISldWorks::GetOpenFileName Method](#)
[ISldWorks::IGetFirstDocument2 Method](#)
[ISldWorks::IGetOpenDocumentByName2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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IActiveDoc Property (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IActiveDoc Property (ISldWorks)

Obsolete. Superseded by [ISldWorks::IActiveDoc2](#).

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property IActiveDoc As ModelDoc

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim value As ModelDoc  
  
value = instance.IActiveDoc
```

C#

```
ModelDoc IActiveDoc {get;}
```

C++/CLI

```
property ModelDoc^ IActiveDoc {  
    ModelDoc^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IActiveDoc](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IActiveDoc2 Property (ISldWorks)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IActiveDoc2 Property (ISldWorks)

Gets the currently active document.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property IActiveDoc2 As ModelDoc2
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim value As ModelDoc2  
  
value = instance.IActiveDoc2
```

C#

```
ModelDoc2 IActiveDoc2 {get;}
```

C++/CLI

```
property ModelDoc2^ IActiveDoc2 {  
    ModelDoc2^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Model document](#) or null if the operation fails

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IActiveDoc2](#).

Example

[Traverse Bodies \(C++\)](#)
[Get Names of Configurations Using Variant \(C++\)](#)
[Get Object's Persistent Reference ID \(C++\)](#)

Remarks

If no document is active, then null is returned. This is a read-only property.

The currently active document cannot be the document that is being edited by the end-user. For example, you can use in-context editing to modify an assembly component. The currently active document is the assembly, but the edit target is the assembly component. For assemblies, you can determine the edit target by using the [IAssemblyDoc::GetEditTarget](#) or [IAssemblyDoc::IGetEditTarget2](#) method.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::ActiveDoc Property](#)
[ISldWorks::ActivateDoc2 Method](#)
[ISldWorks::EnumDocuments2 Method](#)
[ISldWorks::GetFirstDocument Method](#)
[ISldWorks::GetOpenDocument Method](#)
[ISldWorks::GetOpenDocumentByName Method](#)
[ISldWorks::GetOpenedFileInfo Method](#)
[ISldWorks::GetOpenFileName Method](#)
[ISldWorks::IActivateDoc3 Method](#)
[ISldWorks::IGetOpenDocumentByName2 Method](#)
[ISldWorks::OpenDoc6 Method](#)
[ISldWorks::IGetFirstDocument2 Method](#)

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ICopyDocument Method (ISldWorks)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : ICopyDocument Method (ISldWorks)

Copies a document and optionally updates references to it.

.NET Syntax

Visual Basic (Declaration)

```
Function ICopyDocument( _  
    ByVal SourceDoc As System.String, _  
    ByVal DestDoc As System.String, _  
    ByVal ChildCount As System.Integer, _  
    ByRef FromChildren As System.String, _  
    ByRef ToChildren As System.String, _  
    ByVal Option As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim SourceDoc As System.String  
Dim DestDoc As System.String  
Dim ChildCount As System.Integer  
Dim FromChildren As System.String  
Dim ToChildren As System.String  
Dim Option As System.Integer  
Dim value As System.Integer  
  
value = instance.ICopyDocument(SourceDoc, DestDoc, ChildCount, FromChildren, ToChildren, Option)
```

C#

```
System.int ICopyDocument(  
    System.string SourceDoc,  
    System.string DestDoc,  
    System.int ChildCount,  
    ref System.string FromChildren,  
    ref System.string ToChildren,  
    System.int Option  
)
```

C++/CLI

```
System.int ICopyDocument(  
& System.String^ SourceDoc,  
& System.String^ DestDoc,  
& System.int ChildCount,  
& System.String^% FromChildren,  
& System.String^% ToChildren,  
& System.int Option  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SourceDoc

Full path and filename of the document to copy

DestDoc

Full path and filename of the document to which to copy SourceDoc

ChildCount

Number of child documents for SourceDoc

FromChildren

Array containing the full path and filenames of the child documents dependent on the document specified for SourceDoc

ToChildren

Array of strings containing the new full path and filenames of the child documents to which to copy the child documents specified for FromChildren

Option

Copy options as defined by [swMoveCopyOptions_e](#)

Return Value

Success or error code as defined by [swMoveCopyError_e](#)

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::ICopyDocument](#).

Example

[Copy_Document \(C++\)](#)

Remarks

There can be no documents open when using this method.

See Also

[ISldWorks Interface](#)

[ISldWorks Members](#)

[ISldWorks::MoveDocument Method](#)

[IModelDocExtension::SaveAs Method](#)

[ISldWorks::CopyDocument Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

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ICreatePropertyManagerPage Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : ICreatePropertyManagerPage Method (ISldWorks)

Creates a PropertyManager page.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePropertyManagerPage( _  
    ByVal Title As System.String, _  
    ByVal Options As System.Integer, _  
    ByVal Handler As System.Object, _  
    ByRef Errors As System.Integer _  
    ) As PropertyManagerPage2
```

Visual Basic (Usage)

```
Dim instance As ISldWorks  
Dim Title As System.String  
Dim Options As System.Integer  
Dim Handler As System.Object  
Dim Errors As System.Integer  
Dim value As PropertyManagerPage2
```

```
value = instance.ICreatePropertyManagerPage(Title, Options, Handler, Errors)
```

C#



```
PropertyManagerPage2 ICreatePropertyManagerPage(  
    System.string Title,  
    System.int Options,  
    System.object Handler,  
    out System.int Errors  
)
```

C++/CLI

```
PropertyManagerPage2^ ICreatePropertyManagerPage(  
& System.String^ Title,  
& System.int Options,  
& System.Object^ Handler,  
& [Out] System.int Errors  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Title

Title of the page

Options

Options as defined in [swPropertyManagerPageOptions_e](#)

Handler

Pointer to the event handler for this page ([IPropertyManagerPage2Handler5](#))

Errors

Status of the creation as defined in [swPropertyManagerPageStatus_e](#)

Return Value

[PropertyManager page](#)

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::ICreatePropertyManagerPage](#).

Remarks

Specify the Locked option in the Options parameter when you create your PropertyManager page. It is important that when a handler (such as [IPropertyManagerPage2Handler5::OnButtonPress](#) or [IPropertyManagerPage2Handler5::OnClose](#)) is finished and control returns to SOLIDWORKS that the PropertyManager page is still there. If the PropertyManager page is not there, SOLIDWORKS might crash. Some methods try to close the PropertyManager page, but you can avoid this scenario by creating the PropertyManager page as Locked.

See Also

[ISldWorks Interface](#)

[ISldWorks Members](#)

[ISldWorks::CreatePropertyManagerPage Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IDefineAttribute Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IDefineAttribute Method (ISldWorks)

Creates an attribute definition, which is the first step in generating attributes.

.NET Syntax

Visual Basic (Declaration)

```
Function IDefineAttribute( _  
    ByVal Name As System.String _  
) As AttributeDef
```

```
Dim instance As ISldWorks
Dim Name As System.String
Dim value As AttributeDef

value = instance.IDefineAttribute(Name)
```

C#

```
AttributeDef IDefineAttribute(
    System.string Name
)
```

C++/CLI

```
AttributeDef^ IDefineAttribute(
    & System.String^ Name
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Name to give to the attribute definition; the name must be unique and qualified among the attributes defined in the current session (see **Remarks**)

Return Value

[Attribute definition](#)

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IDefineAttribute](#).

Remarks

An attribute definition is a container that can hold a group of parameters. For example, you can use this container to hold machining information or BOM data. Instances of this container may then be created as [IAtribute](#) objects on the document object, or on faces, edges, vertices, and features within the document. Once created, you can get any of the [IAtribute](#) objects and query or change its parameter values.

The Name argument used for this attribute definition must be unique and qualified among the attributes defined in the current session. To ensure this, the first three letters of the name must be an officially recognized third-party identifier. Contact SOLIDWORKS Corporation if you need one. You can also use the prefix: "pub" (for "public") if you are just testing.

General sequence of steps in attribute creation

1. Create an attribute definition ([ISldWorks::DefineAttribute](#) or [ISldWorks::IDefineAttribute](#))
2. Add parameters to the attribute definition ([IAtributeDef::AddParameter](#))
3. Register the attribute definition ([IAtributeDef::Register](#))
4. Create instances of the attribute definition on objects throughout the model ([IAtributeDef::CreateInstance5](#))

Perform steps 1 through 3 only once for each working session. In other words, perform steps 1 through 3 when your DLL or EXE is initially loaded or run. Until the DLL is unloaded, or the EXE is closed, you can create as many instances of the attribute definition as you want as shown in step 4.

Calling this method attaches to an existing attribute definition with the specified Name or creates new attribute definition if a match is not found. The attribute definition exists throughout the current SOLIDWORKS Design session and persistent across SOLIDWORKS Design sessions if an attribute instance with that definition exists in the document. If an attribute definition exists, you cannot add or remove its parameters.

After an attribute instance is added to a face, edge, vertex, loop, feature, or document, there are several ways to get back to the attribute for query or modification.

1. Because an attribute instance is a feature, you can use any of the standard feature traversal routines (the FeatureByName methods, [IModelDoc2::FirstFeature](#) or [IModelDoc2::!FirstFeature](#), and [IFeature::GetNextFeature](#) or [IModelDoc2::!GetNextFeature](#)) to grab the feature representation of the attribute. After you have the [IFeature](#) object that represents the attribute instance, you can call [IFeature::GetSpecificFeature2](#) to get the underlying [IAtribute](#) object.
2. If the end-user selected the attribute in the FeatureManager design tree, you can use standard selection control to get the [IFeature](#) object representing the attribute instance ([ISelectionMgr::GetSelectedObject6](#)) and then call [IFeature::GetSpecificFeature2](#) to get the underlying [IAtribute](#) object.
3. If you are traversing body topology, then you can locate any [IAtribute](#) objects with a call to [IEntity::FindAttribute](#) or [IEntity::!FindAttribute](#) from the particular [IEntity](#) object (the is, face, edge, and so on). If the attribute instance was placed on the document, then you would need to use option 1 to get back to it.

After you have the [IAtribute](#) object that you want to query or modify, use the [IAtribute::GetParameter](#) or [IAtribute::!GetParameter](#) method to return the desired attribute parameter object. The [IParameter](#) object provides a set of functions to query or modify specific attribute values.

See Also

[ISldWorks interface](#)
[ISldWorks Members](#)
[ISldWorks::DefineAttribute Method](#)

IEnableStereoDisplay Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IEnableStereoDisplay Method (ISldWorks)

Obsolete and not superseded. Functionality no longer implemented.

.NET Syntax

Visual Basic (Declaration)

```
Function IEnableStereoDisplay( _
    ByVal BEnable As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISldWorks
Dim BEnable As System.Boolean
Dim value As System.Boolean
```

```
value = instance.IEnableStereoDisplay(BEnable)
```

C#

```
System.bool IEnableStereoDisplay(
    System.bool BEnable
)
```

C++/CLI

```
System.bool IEnableStereoDisplay(
    & System.bool BEnable
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BEnable

Bitfield flags that control the application's stereo support status

Return Value

True if stereo attribute setting was successful, false if not

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IEnableStereoDisplay](#).

Remarks

You must select the **Enable Quadbuffered Stereo API** check box for NVIDIA Quadro4 OpenGL video cards before opening any windows.

Calling this method with bit #0, the low bit, of the flags field set (flags |= 0x01) sets up all subsequently opened document windows to be capable of displaying stereo; already opened document windows are not affected. If this method is not called, or if it is called with the low bit of flags cleared, the document windows are not be capable of displaying stereo.

Bit #1 controls whether or not 3D scenes are displayed to separate left-eye and right-eye stereoscopic buffers. If bit #1 of the flags field is set (flags |= 0x02), separately buffered stereo pairs are rendered for all windows with 3D scenes that are capable of displaying stereo.

Thus, any plug-in that might need to display stereoscopically should, upon loading, immediately call this method, with the flags field set to 0x01. To actually activate stereo display, call this method with the flags field set to 0x03. To deactivate stereo display, call this method with the flags field set to 0x01. Finally, just before the plug-in uninstalls, restore SOLIDWORKS' default behavior by calling this method with the flags field set to 0x00.

For the stereo pair renderings to appear stereoscopic, call [IModelView::SetStereoSeparation](#) or [IModelView::ISetStereoSeparation](#). By default, the stereo rendering-control attributes that are set by IModelView::SetStereoSeparation or IModelView::ISetStereoSeparation are set for identical left-eye and right-eye renderings.

See Also

IGetColorTable Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetColorTable Method (ISldWorks)

Gets a color table from the SOLIDWORKS Design application.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetColorTable() As ColorTable
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim value As ColorTable  
  
value = instance.IGetColorTable()
```

C#

```
ColorTable IGetColorTable()
```

C++/CLI

```
ColorTable^ IGetColorTable();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Color table](#)

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetColorTable](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetColorTable Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

IGetConfigurationNames Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetConfigurationNames Method (ISldWorks)

Gets the names of the configuration in this SOLIDWORKS document, whether the document is opened or closed.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetConfigurationNames( _  
    ByVal FilePathName As System.String, _  
    ByVal Count As System.Integer _  
) As System.String
```

Visual Basic (Usage)



```

Dim instance As ISldWorks
Dim FilePathName As System.String
Dim Count As System.Integer
Dim value As System.String

value = instance.IGetConfigurationNames(FilePathName, Count)

```

C#

```

System.string IGetConfigurationNames(
    System.string FilePathName,
    System.int Count
)

```

C++/CLI

```

System.String^ IGetConfigurationNames(
    & System.String^ FilePathName,
    & System.int Count
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FilePathName

Path and file name for the SOLIDWORKS document

Count

Number of configurations

Return Value

- in-process, unmanaged C++: Pointer to an array of strings containing the names of the configurations in this SOLIDWORKS document
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISldWorks::GetConfigurationCount](#) to get the value for Count.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetActiveConfigurationName](#)
[ISldWorks::GetConfigurationNames Method](#)
[IModelDoc2::GetConfigurationNames](#)
[IModelDoc2::IGetConfigurationNames](#)
[IConfiguration Interface](#)
[IConfigurationManager Interface](#)

Availability

SOLIDWORKS 2003 SP1, Revision Number 11.1

SOLIDWORKS API Help

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IGetDocumentDependencies Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetDocumentDependencies Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::IGetDocumentDependencies2](#).

.NET Syntax

Visual Basic (Declaration)

```

Function IGetDocumentDependencies( _
    ByVal Document As System.String, _
    ByVal Traverseflag As System.Integer, _
    ByVal Searchflag As System.Integer _
) As System.String

```

```

Dim instance As ISldWorks
Dim Document As System.String
Dim Traverseflag As System.Integer
Dim Searchflag As System.Integer
Dim value As System.String

value = instance.IGetDocumentDependencies(Document, Traverseflag, Searchflag)

```

C#

```

System.string IGetDocumentDependencies(
    System.string Document,
    System.int Traverseflag,
    System.int Searchflag
)

```

C++/CLI

```

System.String^ IGetDocumentDependencies(
    & System.String^ Document,
    & System.int Traverseflag,
    & System.int Searchflag
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Document

Traverseflag

Searchflag

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetDocumentDependencies](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetDocumentDependencies2 Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetDocumentDependencies2 Method (ISldWorks)

Gets all of the model dependencies for a document.

.NET Syntax

Visual Basic (Declaration)

```

Function IGetDocumentDependencies2( _
    ByVal Document As System.String, _
    ByVal Traverseflag As System.Boolean, _
    ByVal Searchflag As System.Boolean, _
    ByVal AddReadOnlyInfo As System.Boolean _
) As System.String

```

Visual Basic (Usage)

Copy Code

```

Dim instance As ISldWorks
Dim Document As System.String
Dim Traverseflag As System.Boolean
Dim Searchflag As System.Boolean

```

```
Dim AddReadOnlyInfo As System.Boolean
Dim value As System.String
```

```
value = instance.IGetDocumentDependencies2(Document, Traverseflag, Searchflag, AddReadOnlyInfo)
```

C#

```
System.string IGetDocumentDependencies2(
    System.string Document,
    System.bool Traverseflag,
    System.bool Searchflag,
    System.bool AddReadOnlyInfo
)
```

C++/CLI

```
System.String^ IGetDocumentDependencies2(
    & System.String^ Document,
    & System.bool Traverseflag,
    & System.bool Searchflag,
    & System.bool AddReadOnlyInfo
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Document

Name of the document

Traverseflag

True if you want to traverse down into all dependent files, false if you want only the highest level within the dependencies

Searchflag

Set this argument to True if you want to use the search rules to find dependencies, false looks where the documents were last saved

AddReadOnlyInfo

True if you wish to have read-only information with the filenames; false if not

Return Value

- in-process, unmanaged C++: Pointer to an array of strings with two strings for each document returned in this list of dependent files:
 - File name
 - Filename with the complete pathname; this combination repeats itself for each dependent file found for this document
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISldWorks::IGetDocumentDependenciesCount2](#) to determine the size of the returned array

As an example, calling this method on a drawing document returns a list of all the part and assemblies used in the drawing. If you set the Traverseflag to True, then each of the parts within the assembly file are also returned in this array of strings.

Another example would be calling this method with a derived mirror part. Because a derived mirror part is generated from another part, the list of model dependencies returned by this method would include the original part used to generate the derived mirror part.

Be aware that Library features are completely unassociated. They do not require the library nor do they update when changes are made to the Library feature. Therefore, this method would not return Library features.

If you use this method with an assembly that contains two documents, Part1 and SubAssem1, an example of what might be returned is:

```
[ "Part1", "C:\temp\Part1.SLDPRT", "SubAssem1", "c:\temp\SubAssem1.SLDASM" ]
```

If Traverseflag is set to True, then all of the documents contained in SubAssem1.ASM are also returned. Suppressed components are recognized and returned by this method as a dependent file.

If the AddReadOnlyInfo flag is set to True, then a string of "Read-Only" is added to the list of strings. If Part1 from were read-only, then the resulting set of strings would be:

```
[ "Part1", "C:\temp\Part1.SLDPRT", "Read-Only", "SubAssem1", "c:\temp\SubAssem1.SLDASM", "" ]
```

NOTE: If the SearchFlag is set to True, then the current directory is set to the directory of the document file. This replicates the interactive behavior of the **References** button in the File Open dialog window.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetDocumentDependencies2 Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDocumentDependenciesCount2 Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetDocumentDependenciesCount2 Method (ISldWorks)

Gets the size of the array needed for a call to [ISldWorks::IGetDocumentDependencies2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDocumentDependenciesCount2( _  
    ByVal Document As System.String, _  
    ByVal Traverseflag As System.Boolean, _  
    ByVal Searchflag As System.Boolean, _  
    ByVal AddReadOnlyInfo As System.Boolean _  
    ) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim Document As System.String  
Dim Traverseflag As System.Boolean  
Dim Searchflag As System.Boolean  
Dim AddReadOnlyInfo As System.Boolean  
Dim value As System.Integer  
  
value = instance.IGetDocumentDependenciesCount2(Document, Traverseflag, Searchflag,  
AddReadOnlyInfo)
```

C#

```
System.int IGetDocumentDependenciesCount2(  
    System.string Document,  
    System.bool Traverseflag,  
    System.bool Searchflag,  
    System.bool AddReadOnlyInfo  
)
```

C++/CLI

```
System.int IGetDocumentDependenciesCount2(  
& System.String^ Document,  
& System.bool Traverseflag,  
& System.bool Searchflag,  
& System.bool AddReadOnlyInfo  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Document

Name of the document

Traverseflag

True if you want to traverse down into all dependent files, false if you want only the highest level within the dependencies

Searchflag

Set this argument to True if you want to use the search rules to find dependencies, false looks where the documents were last saved

AddReadOnlyInfo

True if you want to have read-only information with the filenames, false if not

Return Value

Number of strings returned by [ISldWorks::IGetDocumentDependencies2](#)

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetDocumentDependenciesCount2](#).

Remarks

If SearchFlag is set to True, then the current directory is set to the directory of the document file. This is the same as interactively clicking the **References** button in the File Open dialog.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::IGetDocumentDependencies2 Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDocuments Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetDocuments Method (ISldWorks)

Gets the open documents in this SOLIDWORKS Design session.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDocuments(  
    ByVal NumDocuments As System.Integer _  
) As ModelDoc2
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim NumDocuments As System.Integer  
Dim value As ModelDoc2  
  
value = instance.IGetDocuments(NumDocuments)
```

C#

```
ModelDoc2 IGetDocuments(  
    System.int NumDocuments  
)
```

C++/CLI

```
ModelDoc2^ IGetDocuments(  
&    System.int NumDocuments  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumDocuments
Number of open documents

Return Value

- in-process, unmanaged C++: Pointer to an array of open [documents](#) in this SOLIDWORKS Design session
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISldWorks::GetDocumentCount](#) to get NumDocuments.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::CloseAllDocuments Method](#)
[ISldWorks::CloseDoc Method](#)
[ISldWorks::GetFirstDocument Method](#)
[ISldWorks::GetOpenDocument Method](#)
[ISldWorks::GetOpenDocumentByName Method](#)
[ISldWorks::GetDocuments Method](#)
[ISldWorks::IGetFirstDocument2 Method](#)
[ISldWorks::IGetOpenDocumentByName2 Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetEnvironment Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetEnvironment Method (ISldWorks)

Gets the [IEnvironment](#) object.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEnvironment() As Environment
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim value As Environment
```

```
value = instance.IGetEnvironment()
```

C#

```
Environment IGetEnvironment()
```

C++/CLI

```
Environment^ IGetEnvironment();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IEnvironment](#) object

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetEnvironment](#).

Remarks

All numeric values returned from the IEnvironment object are relative to a unit text height of 1.0; i.e., if a symbol has a text height of 0.15, then you the numeric values returned by 0.15.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetEnvironment Method](#)

IGetFirstDocument Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetFirstDocument Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::IGetFirstDocument2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstDocument() As ModelDoc
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim value As ModelDoc  
  
value = instance.IGetFirstDocument()
```

C#

```
ModelDoc IGetFirstDocument()
```

C++/CLI

```
ModelDoc^ IGetFirstDocument();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetFirstDocument](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

IGetFirstDocument2 Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetFirstDocument2 Method (ISldWorks)

Gets the document opened first in this SOLIDWORKS Design session.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstDocument2() As ModelDoc2
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim value As ModelDoc2  
  
value = instance.IGetFirstDocument2()
```

C#

```
ModelDoc2 IGetFirstDocument2()
```

C++/CLI

```
ModelDoc2^ IGetFirstDocument2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Document](#) first opened

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IGetFirstDocument2](#).

Remarks

If the document first opened is an assembly, then this method returns the name of the first component document opened in that assembly.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetFirstDocument Method](#)
[ISldWorks::ActiveDoc Property](#)
[ISldWorks::!ActiveDoc2 Property](#)
[ISldWorks::EnumDocuments2 Method](#)
[ISldWorks::GetOpenDocument Method](#)
[ISldWorks::GetOpenDocumentByName Method](#)
[ISldWorks::CloseAllDocuments Method](#)
[ISldWorks::CloseDoc Method](#)

Availability

SOLIDWORKS 98Plus, datecode 1998319

SOLIDWORKS API Help

[Send comments on this topic](#).

IGetMassProperties Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetMassProperties Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::GetMassProperties2](#) and [ISldWorks::IGetMassProperties2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMassProperties( _  
    ByVal FilePathName As System.String, _  
    ByVal ConfigurationName As System.String, _  
    ByRef MPropsData As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim FilePathName As System.String  
Dim ConfigurationName As System.String  
Dim MPropsData As System.Double  
Dim value As System.Boolean  
  
value = instance.IGetMassProperties(FilePathName, ConfigurationName, MPropsData)
```

C#

```
System.bool IGetMassProperties(  
    System.string FilePathName,  
    System.string ConfigurationName,  
    ref System.double MPropsData  
)
```

C++/CLI

```
System.bool IGetMassProperties(  
& System.String^ FilePathName,  
& System.String^ ConfigurationName,
```

```
& System.double% MPropsData
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FilePathName

ConfigurationName

MPropsData

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetMassProperties](#).

See Also

[ISldWorks Interface](#)

[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetMassProperties2 Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetMassProperties2 Method (ISldWorks)

Gets the mass properties from the specified document for the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMassProperties2( _
    ByVal FilePathName As System.String, _
    ByVal ConfigurationName As System.String, _
    ByRef MPropsData As System.Double, _
    ByVal Accuracy As System.Integer _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISldWorks
Dim FilePathName As System.String
Dim ConfigurationName As System.String
Dim MPropsData As System.Double
Dim Accuracy As System.Integer
Dim value As System.Boolean

value = instance.IGetMassProperties2(FilePathName, ConfigurationName, MPropsData, Accuracy)
```

C#

```
System.bool IGetMassProperties2(
    System.string FilePathName,
    System.string ConfigurationName,
    ref System.double MPropsData,
    System.int Accuracy
)
```

C++/CLI

```
System.bool IGetMassProperties2(
    & System.String^ FilePathName,
    & System.String^ ConfigurationName,
    & System.double% MPropsData,
    & System.int Accuracy
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FilePathName

Document path and file name

ConfigurationName

Name of configuration to use

MPropsData

Array of doubles

Accuracy

- 0 = as is
- 1 = default
- 2 = maximum

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles of size 13; last element is the accuracy at which mass properties are calculated
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

You can get the density of the current SOLIDWORKS part using [ISldWorks::GetUserPreferenceDoubleValue](#) and `swMaterialPropertyDensity`. If the density has not been explicitly set by the user, then SOLIDWORKS uses 1.0. You can also derive the density of the body from the following calculation:

$$\text{Density} = (\text{Mass} / \text{Volume})$$

Consistent with all other functions, this method returns metric units unless otherwise specified.

If this [IModelDoc2](#) object is an assembly, then any suppressed components are not included in the mass property analysis. See [IComponent2::GetSuppression](#) to determine the state of each of the assembly components.

This method returns an empty array if mass properties are not calculated when saving the model. This is a system-level setting and is controlled by [ISldWorks::SetUserPreferenceToggle](#) and `swUpdateMassPropsDuringSave`.

NOTE: The calculated origin for the return values is based on the default coordinate systems of the `IModelDoc2`, not on the a selected coordinate system.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[IMassProperty Interface](#)
[ISldWorks::GetMassProperties2 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMaterialDatabases Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetMaterialDatabases Method (ISldWorks)

Gets the names of the material databases.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMaterialDatabases( _  
    ByVal Count As System.Integer _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As ISldWorks
Dim Count As System.Integer
Dim value As System.String

value = instance.IGetMaterialDatabases(Count)
```

C#

```
System.string IGetMaterialDatabases(
    System.int Count
)
```

C++/CLI

```
System.String^ IGetMaterialDatabases(
    & System.int Count
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of material databases

Return Value

- in-process, unmanaged C++: Pointer to the an array of the names of the material databases
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISldWorks::GetMaterialDatabaseCount](#) to get the value for Count.

Material database names must be unique. Do not re-use the name of a material database.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetMaterialDatabases Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMathUtility Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetMathUtility Method (ISldWorks)

Gets the [IMathUtility](#) interface.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMathUtility() As MathUtility
```

Visual Basic (Usage)



```
Dim instance As ISldWorks
Dim value As MathUtility
```

```
value = instance.IGetMathUtility()
```

C#

```
MathUtility IGetMathUtility()
```

C++/CLI

```
MathUtility^ IGetMathUtility();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IMathUtility](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetMathUtility](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetMathUtility Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetModeler Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetModeler Method (ISldWorks)

Gets the [IModeler](#) interface.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetModeler() As Modeler
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim value As Modeler  
  
value = instance.IGetModeler()
```

C#

```
Modeler IGetModeler()
```

C++/CLI

```
Modeler^ IGetModeler();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IModeler](#)

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetModeler](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

IGetOpenDocumentByName Method (ISldWorks)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetOpenDocumentByName Method (ISldWorks)**Obsolete. Superseded** [ISldWorks::IGetOpenDocumentByName2.](#)

.NET Syntax

Visual Basic (Declaration)

```
Function IGetOpenDocumentByName( _  
    ByVal DocumentName As System.String _  
) As ModelDoc
```

Visual Basic (Usage)

```
Dim instance As ISldWorks  
Dim DocumentName As System.String  
Dim value As ModelDoc  
  
value = instance.IGetOpenDocumentByName(DocumentName)
```

C#

```
ModelDoc IGetOpenDocumentByName(  
    System.string DocumentName  
)
```

C++/CLI

```
ModelDoc^ IGetOpenDocumentByName(  
& System.String^ DocumentName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code.](#)

Parameters

DocumentName

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetOpenDocumentByName.](#)

See Also

[ISldWorks Interface](#)[ISldWorks Members](#)

IGetOpenDocumentByName2 Method (ISldWorks)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetOpenDocumentByName2 Method (ISldWorks)

Gets the open document with the specified name.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetOpenDocumentByName2( _  
    ByVal DocumentName As System.String _  
) As ModelDoc2
```

Visual Basic (Usage)



```
Dim instance As ISldWorks
Dim DocumentName As System.String
Dim value As ModelDoc2

value = instance.IGetOpenDocumentByName2(DocumentName)
```

C#

```
ModelDoc2 IGetOpenDocumentByName2(
    System.string DocumentName
)
```

C++/CLI

```
ModelDoc2^ IGetOpenDocumentByName2(
    & System.String^ DocumentName
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DocumentName

Fully qualified name of the open document (path, filename, and file extension)

Return Value

Open [model document](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetOpenDocumentByName2](#).

Remarks

Only call this method for a document that has already been opened in its own window. If you call this method for a document that has not been opened in its own window, then methods on the returned object may not work as expected.

This method is useful for getting a SOLIDWORKS document object when you only have the document name. For example, SOLIDWORKS ever [FileOpenNotify2](#) provides the filename of the document opened, and you can use that name with this method to get the document's object pointer.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::CloseAllDocuments Method](#)
[ISldWorks::CloseDoc Method](#)
[ISldWorks::GetOpenDocument Method](#)
[ISldWorks::GetOpenDocumentByName Method](#)
[ISldWorks::EnumDocuments2 Method](#)
[ISldWorks::GetFirstDocument Method](#)
[ISldWorks::IGetFirstDocument2 Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetRayTraceRenderer Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetRayTraceRenderer Method (ISldWorks)

Get a ray-trace rendering engine.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetRayTraceRenderer( _
    ByVal RendererType As System.Integer _
```

) As RayTraceRenderer

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim RendererType As System.Integer  
Dim value As RayTraceRenderer  
  
value = instance.IGetRayTraceRenderer(RendererType)
```

C#

```
RayTraceRenderer IGetRayTraceRenderer(  
    System.int RendererType  
)
```

C++/CLI

```
RayTraceRenderer^ IGetRayTraceRenderer(  
& System.int RendererType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

RendererType

Type of ray-trace rendering engine as defined in [swRayTraceRenderType_e](#)

Return Value

[Ray trace renderer](#)

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetRayTraceRenderer](#).

See Also

[ISldWorks Interface](#)

[ISldWorks Members](#)

[ISldWorks::IGetRayTraceRenderer Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSelectionFilters Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetSelectionFilters Method (ISldWorks)

Gets all active selection filters.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSelectionFilters() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim value As System.Integer  
  
value = instance.IGetSelectionFilters()
```

C#

```
System.int IGetSelectionFilters()
```

C++/CLI

```
System.int IGetSelectionFilters();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of long values representing the [swSelectType_e](#) enumeration
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IGetSelectionFilters](#).

Remarks

To determine...	Call...
Size of the array to pass	ISldWorks::IGetSelectionFiltersCount
If the selection filter is active or not	ISldWorks::GetApplySelectionFilter

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetSelectionFilter Method](#)
[ISldWorks::GetSelectionFilters Method](#)
[ISldWorks::SetSelectionFilters Method](#)
[ISldWorks::SetSelectionFilter Method](#)
[ISldWorks::ISetSelectionFilters Method](#)
[ISldWorks::SetApplySelectionFilter Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetSelectionFiltersCount Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetSelectionFiltersCount Method (ISldWorks)

Gets the number of active selection filters.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetSelectionFiltersCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim value As System.Integer
```

```
value = instance.IGetSelectionFiltersCount()
```

C#

```
System.int IGetSelectionFiltersCount()
```

C++/CLI

```
System.int IGetSelectionFiltersCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of all active selection filters

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IGetSelectionFiltersCount](#).

Remarks

Call this method before calling [ISldWorks::IGetSelectionFilters](#) to determine the size of the array to pass that to that method.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetSelectionFilter Method](#)
[ISldWorks::GetSelectionFilters Method](#)
[ISldWorks::IGetSelectionFilters Method](#)
[ISldWorks::SetSelectionFilter Method](#)
[ISldWorks::SetSelectionFilters Method](#)
[ISldWorks::GetApplySelectionFilter Method](#)
[ISldWorks::SetApplySelectionFilter Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTemplateSizes Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetTemplateSizes Method (ISldWorks)

Gets the sheet properties from a template document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTemplateSizes( _  
    ByVal FileName As System.String, _  
    ByRef PaperSize As System.Integer, _  
    ByRef Width As System.Double, _  
    ByRef Height As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim FileName As System.String  
Dim PaperSize As System.Integer  
Dim Width As System.Double  
Dim Height As System.Double  
Dim value As System.Boolean  
  
value = instance.IGetTemplateSizes(FileName, PaperSize, Width, Height)
```

C#

```
System.bool IGetTemplateSizes(
    System.string FileName,
    out System.int PaperSize,
    out System.double Width,
    out System.double Height
)
```

C++/CLI

```
System.bool IGetTemplateSizes(
    & System.String^ FileName,
    & [Out] System.int PaperSize,
    & [Out] System.double Width,
    & [Out] System.double Height
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Name of template with full directory path

PaperSize

Paper size as defined in [swDwgPaperSizes_e](#)

Width

Custom paper width if PaperSize is set to swDwgPapersUserDefined

Height

Custom paper height if PaperSize is set to swDwgPapersUserDefined

Return Value

True if the operation is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IGetTemplateSizes](#).

See Also

[ISldWorks Interface](#)

[ISldWorks Members](#)

[ISldWorks::GetTemplateSizes Method](#)

[ISldWorks::GetDocumentTemplate Method](#)

[ISldWorks::PreSelectDwgTemplateSize Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetUserTypeLibReferences Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetUserTypeLibReferences Method (ISldWorks)

Gets the specified user-specified type library references.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetUserTypeLibReferences( _
    ByVal NCount As System.Integer _
) As System.String
```

Visual Basic (Usage)

```
Dim instance As ISldWorks
Dim NCount As System.Integer
Dim value As System.String
```



```
value = instance.IGetUserTypeLibReferences(NCount)
```

C#

```
System.string IGetUserTypeLibReferences(  
    System.int NCount  
)
```

C++/CLI

```
System.String^ IGetUserTypeLibReferences(  
& System.int NCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NCount

Number of user-specified type libraries

Return Value

- in-process, unmanaged C++: Pointer to an array of strings for the user-specified type library references.
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISldWorks::GetUserTypeLibReferenceCount](#) to get the value for NCount.

See [Type Libraries](#) for details.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::ISetUserTypeLibReferences Method](#)
[ISldWorks::RemoveUserTypeLibReferences Method](#)
[ISldWorks::UserTypeLibReferences Property](#)
[ISldWorks::IRemoveUserTypeLibReferences Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetUserUnit Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetUserUnit Method (ISldWorks)

Gets an empty [IUserUnit](#) object of the specified type.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetUserUnit( _  
    ByVal UnitType As System.Integer _  
) As IUserUnit
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim UnitType As System.Integer  
Dim value As IUserUnit
```

```
value = instance.IGetUserUnit(UnitType)
```

C#

```
UserUnit IGetUserUnit(  
    System.int UnitType  
)
```

C++/CLI

```
UserUnit^ IGetUserUnit(  
& System.int UnitType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UnitType

Unit type as defined in [swUserUnitsType_e](#)

Return Value

IUserUnit

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetUserUnit](#).

Remarks

The instance of IUserUnit returned by this method is read-write, not persistent, and not tied to any document. Use this instance as a template to store units properties at runtime.

Call [IModelDoc2::IGetUserUnit](#) to get the user units of a document.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetUserUnit Method](#)
[IModelDoc2::GetUserUnit Method](#)

Availability

SOLIDWORKS 2001 SP1, Revision Number 9.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetVersionHistoryCount Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IGetVersionHistoryCount Method (ISldWorks)

Gets the size of the array required to hold data returned by [ISldWorks::IVersionHistory](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetVersionHistoryCount( _  
    ByVal FileName As System.String _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim FileName As System.String  
Dim value As System.Integer
```

```
value = instance.IGetVersionHistoryCount(FileName)
```

C#

```
System.int IGetVersionHistoryCount(  
    System.string FileName  
)
```

C++/CLI

```
System.int IGetVersionHistoryCount(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Full pathname of the model for which to get the version history

Return Value

Size of array required for [ISldWorks::IVersionHistory](#)

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IGetVersionHistoryCount](#).

Remarks

If the file is not found or an error occurs, then this method returns 0.

To get the version history of a document that is currently open, use [IModelDoc2::VersionHistory](#) or [IModelDoc2::IVersionHistory](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::VersionHistory Method](#)

Availability

SOLIDWORKS 98Plus, datecode 1998319

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMoveDocument Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IMoveDocument Method (ISldWorks)

Moves a document and optionally updates references to it.

.NET Syntax

Visual Basic (Declaration)

```
Function IMoveDocument( _  
    ByVal SourceDoc As System.String, _  
    ByVal DestDoc As System.String, _  
    ByVal ChildCount As System.Integer, _  
    ByRef FromChildren As System.String, _  
    ByRef ToChildren As System.String, _  
    ByVal Option As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim SourceDoc As System.String  
Dim DestDoc As System.String  
Dim ChildCount As System.Integer
```

```
Dim FromChildren As System.String
Dim ToChildren As System.String
Dim Option As System.Integer
Dim value As System.Integer
```

```
value = instance.IMoveDocument(SourceDoc, DestDoc, ChildCount, FromChildren, ToChildren, Option)
```

C#

```
System.int IMoveDocument(
    System.string SourceDoc,
    System.string DestDoc,
    System.int ChildCount,
    ref System.string FromChildren,
    ref System.string ToChildren,
    System.int Option
)
```

C++/CLI

```
System.int IMoveDocument(
    & System.String^ SourceDoc,
    & System.String^ DestDoc,
    & System.int ChildCount,
    & System.String^% FromChildren,
    & System.String^% ToChildren,
    & System.int Option
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SourceDoc

Full path and filename of the document to move

DestDoc

Full path and filename of the new document to which to move the document specified for SourceDoc

ChildCount

Number of child documents for SourceDoc

FromChildren

Array of strings containing the full path and filenames of the child documents dependent on the document specified for SourceDoc

ToChildren

Array of strings containing the new full path and filenames for the child documents to which to move the documents specified for FromChildren

Option

Move options as defined by [swMoveCopyOptions_e](#)

Return Value

Success or error code as defined by [swMoveCopyError_e](#)

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IMoveDocument](#).

See Also

[ISldWorks Interface](#)

[ISldWorks Members](#)

[ISldWorks::MoveDocument Method](#)

[ISldWorks::ICopyDocument Method](#)

[ISldWorks::CopyDocument Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ImportHoleWizardItem Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : ImportHoleWizardItem Method (ISldWorks)

Imports data for the specified Hole Wizard standard.

.NET Syntax

Visual Basic (Declaration)

```
Function ImportHoleWizardItem( _  
    ByVal StdToImport As System.String, _  
    ByVal DestinationFilePath As System.String, _  
    ByVal ReplaceData As System.Boolean, _  
    ByVal ErrorFile As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim StdToImport As System.String  
Dim DestinationFilePath As System.String  
Dim ReplaceData As System.Boolean  
Dim ErrorFile As System.Boolean  
Dim value As System.Integer  
  
value = instance.ImportHoleWizardItem(StdToImport, DestinationFilePath, ReplaceData, ErrorFile)
```

C#

```
System.int ImportHoleWizardItem(  
    System.string StdToImport,  
    System.string DestinationFilePath,  
    System.bool ReplaceData,  
    System.bool ErrorFile  
)
```

C++/CLI

```
System.int ImportHoleWizardItem(  
& System.String^ StdToImport,  
& System.String^ DestinationFilePath,  
& System.bool ReplaceData,  
& System.bool ErrorFile  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StdToImport

Standard to import (see **Remarks**)

DestinationFilePath

Path and name of file to import (see **Remarks**)

ReplaceData

True to replace data, false to not

ErrorFile

True to create an error file, false to not

Return Value

0 if the Hole Wizard standard imported successfully, 1 if not

Visual Basic for Applications (VBA) Syntax

See [SldWorks::ImportHoleWizardItem](#).

Example

'VBA Preconditions'

'Open a part.

'Ensure the *.xlsx file exists.

```
Dim swApp As SldWorks.SldWorks  
Dim SourceHWItem As String  
Dim DestinationFoldeName As String  
Dim longstatus As Long
```

Option Explicit

```

Sub Main()

    Set swApp = Application.SldWorks

    SourceHWItem = "ansi inch\Counterbore Holes\Hex Bolt"

    DestinationFolderName = "C:\temp\ANSI Inch_Counterbore Holes_Hex Bolt.xlsx"

    longstatus = swApp.ImportHoleWizardItem(SourceHWItem, DestinationFolderName, True, True)

    Exit Sub

End Sub

```

Remarks

Specify StdToImport with the path and file name as shown on the Hole Wizard tab of the Toolbox, e.g., **"ansi inch\Counterbore Holes\Hex Bolt"**.

Specify DestinationFilePath with the path and file name of the Excel Workbook (*.xlsx) you want to import. The Excel file name is formatted as follows:

standard_hole class_hole type.xlsx

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::ExportHoleWizardItem Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ImportToolboxItem Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : ImportToolboxItem Method (ISldWorks)

Imports data for the specified Toolbox standard.

.NET Syntax

Visual Basic (Declaration)

```

Function ImportToolboxItem( _
    ByVal StdToImport As System.String, _
    ByVal DestinationFilePath As System.String _
) As System.Integer

```

Visual Basic (Usage)



```

Dim instance As ISldWorks
Dim StdToImport As System.String
Dim DestinationFilePath As System.String
Dim value As System.Integer

value = instance.ImportToolboxItem(StdToImport, DestinationFilePath)

```

C#

```
System.int ImportToolboxItem(
    System.string StdToImport,
    System.string DestinationFilePath
)
```

C++/CLI

```
System.int ImportToolboxItem(
    & System.String^ StdToImport,
    & System.String^ DestinationFilePath
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StdToImport

Standard to import (see **Remarks**)

DestinationFilePath

Path and name of file to import (see **Remarks**)

Return Value

0 if the Toolbox standard imported successfully, 1 if not

Visual Basic for Applications (VBA) Syntax

See [SldWorks::ImportToolboxItem](#).

Example

```
'VBA Preconditions'

'Open a part.
'Ensure the *.xlsx file exists.

Dim swApp As SldWorks.SldWorks
Dim SourceHWItem As String
Dim DestinationFolderName As String
Dim longstatus As Long
Option Explicit

Sub Main()

    Set swApp = Application.SldWorks

    SourceHWItem = "ansi inch\bolts and screws\hex head\heavy hex bolt_ai.sldprt"

    DestinationFolderName = "C:\temp\AI_Heavy Hex Bolt.xlsx"

    longstatus = swApp.ImportToolboxItem(SourceHWItem, DestinationFolderName)

    Exit Sub

End Sub
```

Remarks

Specify StdToImport with the path and file name beneath **C:\SOLIDWORKS Datalbrowser**, e.g., "ansi inch\bolts and screws\hex head\heavy hex bolt_ai.sldprt".

Specify DestinationFilePath with the path and file name of the Excel Workbook (*.xlsx) you want to import. The Excel file name is formatted as follows:

standard_hole class_hole type.xlsx

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::ExportToolboxItem Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

INewAssembly Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : INewAssembly Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::NewDocument](#) and [ISldWorks::INewDocument2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function INewAssembly() As AssemblyDoc
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim value As AssemblyDoc  
  
value = instance.INewAssembly()
```

C#

```
AssemblyDoc INewAssembly()
```

C++/CLI

```
AssemblyDoc^ INewAssembly();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SldWorks::INewAssembly](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

INewDocument Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : INewDocument Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::INewDocument2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function INewDocument( _
    ByVal TemplateName As System.String, _
    ByVal PaperSize As System.Integer, _
    ByVal Width As System.Double, _
    ByVal Height As System.Double _
) As ModelDoc
```

Visual Basic (Usage)



```
Dim instance As ISldWorks
Dim TemplateName As System.String
Dim PaperSize As System.Integer
Dim Width As System.Double
Dim Height As System.Double
Dim value As ModelDoc

value = instance.INewDocument(TemplateName, PaperSize, Width, Height)
```

C#

```
ModelDoc INewDocument(
    System.string TemplateName,
    System.int PaperSize,
    System.double Width,
    System.double Height
)
```

C++/CLI

```
ModelDoc^ INewDocument(
    & System.String^ TemplateName,
    & System.int PaperSize,
    & System.double Width,
    & System.double Height
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TemplateName

PaperSize

Width

Height

Visual Basic for Applications (VBA) Syntax

See [SldWorks::INewDocument](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

INewDocument2 Method (ISldWorks)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : INewDocument2 Method (ISldWorks)

Creates a new document based on the specified template.

.NET Syntax

Visual Basic (Declaration)

```
Function INewDocument2( _
    ByVal TemplateName As System.String, _
    ByVal PaperSize As System.Integer, _
    ByVal Width As System.Double, _
    ByVal Height As System.Double _
) As ModelDoc2
```

Visual Basic (Usage)



```
Dim instance As ISldWorks
Dim TemplateName As System.String
Dim PaperSize As System.Integer
Dim Width As System.Double
Dim Height As System.Double
Dim value As ModelDoc2

value = instance.INewDocument2(TemplateName, PaperSize, Width, Height)
```

C#

```
ModelDoc2 INewDocument2(
    System.string TemplateName,
    System.int PaperSize,
    System.double Width,
    System.double Height
)
```

C++/CLI

```
ModelDoc2^ INewDocument2(
    & System.String^ TemplateName,
    & System.int PaperSize,
    & System.double Width,
    & System.double Height
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TemplateName

Name of the template to use for creating the new document

PaperSize

Size of paper as defined in [swDwgPaperSizes_e](#)

Width

Width of paper; used only when PaperSize is swDwgPapersUserDefined

Height

Height of paper; used only when PaperSize is swDwgPapersUserDefined

Return Value

Newly created [document](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [SldWorks::INewDocument2](#).

Example

[Access Assembly \(C++\)](#)

Remarks

To get the default template filename, use [ISldWorks::GetUserPreferenceStringValue](#):

- for parts: swDefaultTemplatePart
- for assemblies: swDefaultTemplateAssembly
- for drawings: swDefaultTemplateDrawing

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetDocumentTemplate Method](#)
[ISldWorks::NewDocument Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

INewDrawing Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : INewDrawing Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::NewDocument](#) and [ISldWorks::INewDocument2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function INewDrawing( _  
    ByVal TemplateToUse As System.Integer _  
) As DrawingDoc
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim TemplateToUse As System.Integer  
Dim value As DrawingDoc  
  
value = instance.INewDrawing(TemplateToUse)
```

C#

```
DrawingDoc INewDrawing(  
    System.int TemplateToUse  
)
```

C++/CLI

```
DrawingDoc^ INewDrawing(  
    & System.int TemplateToUse  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TemplateToUse

Visual Basic for Applications (VBA) Syntax

See [SldWorks::INewDrawing](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

INewDrawing2 Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : INewDrawing2 Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::NewDocument](#) and [ISldWorks::INewDocument2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function INewDrawing2( _  
    ByVal TemplateToUse As System.Integer, _  
    ByVal TemplateName As System.String, _  
    ByVal PaperSize As System.Integer, _  
    ByVal Width As System.Double, _  
    ByVal Height As System.Double _  
) As DrawingDoc
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim TemplateToUse As System.Integer  
Dim TemplateName As System.String  
Dim PaperSize As System.Integer  
Dim Width As System.Double  
Dim Height As System.Double  
Dim value As DrawingDoc  
  
value = instance.INewDrawing2(TemplateToUse, TemplateName, PaperSize, Width, Height)
```

C#

```
DrawingDoc INewDrawing2(  
    System.int TemplateToUse,  
    System.string TemplateName,  
    System.int PaperSize,  
    System.double Width,  
    System.double Height  
)
```

C++/CLI

```
DrawingDoc^ INewDrawing2(  
& System.int TemplateToUse,  
& System.String^ TemplateName,  
& System.int PaperSize,  
& System.double Width,  
& System.double Height  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TemplateToUse

TemplateName

PaperSize

Width

Height

Visual Basic for Applications (VBA) Syntax

See [SldWorks::INewDrawing2](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

Obsolete. Superseded by [ISldWorks::NewDocument](#) and [ISldWorks::INewDocument2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function INewPart() As PartDoc
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim value As PartDoc
```

```
value = instance.INewPart()
```

C#

```
PartDoc INewPart()
```

C++/CLI

```
PartDoc^ INewPart();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SldWorks::INewPart](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InstallQuickTipGuide Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : InstallQuickTipGuide Method (ISldWorks)

Implements your add-in's copy of the [Quick Tips](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InstallQuickTipGuide( _  
    ByVal PInterface As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim PInterface As System.Object
```

```
instance.InstallQuickTipGuide(PInterface)
```

C#

```
void InstallQuickTipGuide(  
    System.object PInterface  
)
```

C++/CLI

```
void InstallQuickTipGuide(
& System.Object^ PInterface
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PInterface
Your add-in's copy of the [Quick Tips](#)

Visual Basic for Applications (VBA) Syntax

See [SldWorks::InstallQuickTipGuide](#).

Remarks

The name of your add-in's Quick Tips is added to the SOLIDWORKS Help menu. See [ISwQuickTip::MenuName](#) for details.

See [Quick Tips and Bubble ToopTips](#) for details.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::RefreshQuickTipWindow Method](#)
[ISldWorks::UninstallQuickTipGuide Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IOpenDoc Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IOpenDoc Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::OpenDoc6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IOpenDoc( _
    ByVal Name As System.String, _
    ByVal Type As System.Integer _
) As ModelDoc
```

Visual Basic (Usage)



```
Dim instance As ISldWorks
Dim Name As System.String
Dim Type As System.Integer
Dim value As ModelDoc
```

```
value = instance.IOpenDoc(Name, Type)
```

C#

```
ModelDoc IOpenDoc(
    System.string Name,
    System.int Type
)
```

C++/CLI

```
ModelDoc^ IOpenDoc(
& System.String^ Name,
```

```
& System.int Type
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Type

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IOpenDoc](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IOpenDoc2 Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IOpenDoc2 Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::OpenDoc6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IOpenDoc2( _
    ByVal FileName As System.String, _
    ByVal Type As System.Integer, _
    ByVal ReadOnly As System.Boolean, _
    ByVal ViewOnly As System.Boolean, _
    ByVal Silent As System.Boolean, _
    ByRef Errors As System.Integer _
) As ModelDoc
```

Visual Basic (Usage)



```
Dim instance As ISldWorks
Dim FileName As System.String
Dim Type As System.Integer
Dim ReadOnly As System.Boolean
Dim ViewOnly As System.Boolean
Dim Silent As System.Boolean
Dim Errors As System.Integer
Dim value As ModelDoc

value = instance.IOpenDoc2(FileName, Type, ReadOnly, ViewOnly, Silent, Errors)
```

C#

```
ModelDoc IOpenDoc2(
    System.string FileName,
    System.int Type,
    System.bool ReadOnly,
    System.bool ViewOnly,
    System.bool Silent,
    out System.int Errors
)
```

C++/CLI

```
ModelDoc^ IOpenDoc2(
    & System.String^ FileName,
    & System.int Type,
```

```
& System.bool ReadOnly,  
& System.bool ViewOnly,  
& System.bool Silent,  
& [Out] System.int Errors  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Type

ReadOnly

ViewOnly

Silent

Errors

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IOpenDoc2](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IOpenDoc3 Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IOpenDoc3 Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::OpenDoc6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IOpenDoc3( _  
    ByVal FileName As System.String, _  
    ByVal Type As System.Integer, _  
    ByVal ReadOnly As System.Boolean, _  
    ByVal ViewOnly As System.Boolean, _  
    ByVal RapidDraft As System.Boolean, _  
    ByVal Silent As System.Boolean, _  
    ByRef Errors As System.Integer _  
) As ModelDoc
```

Visual Basic (Usage)

```
Dim instance As ISldWorks  
Dim FileName As System.String  
Dim Type As System.Integer  
Dim ReadOnly As System.Boolean  
Dim ViewOnly As System.Boolean  
Dim RapidDraft As System.Boolean  
Dim Silent As System.Boolean  
Dim Errors As System.Integer  
Dim value As ModelDoc
```

```
value = instance.IOpenDoc3(FileName, Type, ReadOnly, ViewOnly, RapidDraft, Silent, Errors)
```

C#

```
ModelDoc IOpenDoc3(  
    System.string FileName,  
    System.int Type,  
    System.bool ReadOnly,
```



```
System.bool ViewOnly,  
System.bool RapidDraft,  
System.bool Silent,  
out System.int Errors  
)
```

C++/CLI

```
ModelDoc^ IOpenDoc3(  
& System.String^ FileName,  
& System.int Type,  
& System.bool ReadOnly,  
& System.bool ViewOnly,  
& System.bool RapidDraft,  
& System.bool Silent,  
& [Out] System.int Errors  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Type

ReadOnly

ViewOnly

RapidDraft

Silent

Errors

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IOpenDoc3](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IOpenDoc4 Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IOpenDoc4 Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::OpenDoc6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IOpenDoc4( _  
    ByVal FileName As System.String, _  
    ByVal Type As System.Integer, _  
    ByVal Options As System.Integer, _  
    ByVal Configuration As System.String, _  
    ByRef Errors As System.Integer _  
) As ModelDoc
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim FileName As System.String  
Dim Type As System.Integer  
Dim Options As System.Integer  
Dim Configuration As System.String  
Dim Errors As System.Integer
```

```
Dim value As ModelDoc
```

```
value = instance.IOpenDoc4(FileName, Type, Options, Configuration, Errors)
```

C#

```
ModelDoc IOpenDoc4(  
    System.string FileName,  
    System.int Type,  
    System.int Options,  
    System.string Configuration,  
    out System.int Errors  
)
```

C++/CLI

```
ModelDoc^ IOpenDoc4(  
& System.String^ FileName,  
& System.int Type,  
& System.int Options,  
& System.String^ Configuration,  
& [Out] System.int Errors  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Type

Options

Configuration

Errors

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IOpenDoc4](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IOpenDoc5 Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IOpenDoc5 Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::OpenDoc6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IOpenDoc5( _  
    ByVal FileName As System.String, _  
    ByVal Type As System.Integer, _  
    ByVal Options As System.Integer, _  
    ByVal Configuration As System.String, _  
    ByRef Errors As System.Integer _  
) As ModelDoc2
```

Visual Basic (Usage)

```
Dim instance As ISldWorks  
Dim FileName As System.String  
Dim Type As System.Integer
```



```
Dim Options As System.Integer
Dim Configuration As System.String
Dim Errors As System.Integer
Dim value As ModelDoc2

value = instance.IOpenDoc5(FileName, Type, Options, Configuration, Errors)
```

C#

```
ModelDoc2 IOpenDoc5(
    System.string FileName,
    System.int Type,
    System.int Options,
    System.string Configuration,
    out System.int Errors
)
```

C++/CLI

```
ModelDoc2^ IOpenDoc5(
    & System.String^ FileName,
    & System.int Type,
    & System.int Options,
    & System.String^ Configuration,
    [Out] System.int Errors
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Type

Options

Configuration

Errors

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IOpenDoc5](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IOpenDocSilent Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IOpenDocSilent Method (ISldWorks)

Obsolete. Superseded by [ISldWorks::OpenDoc6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IOpenDocSilent( _
    ByVal FileName As System.String, _
    ByVal Type As System.Integer, _
    ByRef Errors As System.Integer _
) As ModelDoc
```

Visual Basic (Usage)

```
Dim instance As ISldWorks
Dim FileName As System.String
```



```

Dim Type As System.Integer
Dim Errors As System.Integer
Dim value As ModelDoc

value = instance.IOpenDocSilent(FileName, Type, Errors)

```

C#

```

ModelDoc IOpenDocSilent(
    System.string FileName,
    System.int Type,
    out System.int Errors
)

```

C++/CLI

```

ModelDoc^ IOpenDocSilent(
    & System.String^ FileName,
    & System.int Type,
    & [Out] System.int Errors
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Type

Errors

Visual Basic for Applications (VBA) Syntax

See [SldWorks::IOpenDocSilent](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IRemoveUserTypeLibReferences Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IRemoveUserTypeLibReferences Method (ISldWorks)

Removes the user-specified type library references.

.NET Syntax

Visual Basic (Declaration)

```

Function IRemoveUserTypeLibReferences( _
    ByVal NCount As System.Integer, _
    ByRef BstrTlbRef As System.String _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As ISldWorks
Dim NCount As System.Integer
Dim BstrTlbRef As System.String
Dim value As System.Boolean

```

```

value = instance.IRemoveUserTypeLibReferences(NCount, BstrTlbRef)

```

C#

```

System.bool IRemoveUserTypeLibReferences(
    System.int NCount,

```

```
    ref System.string BstrTlbRef
)
```

C++/CLI

```
System.bool IRemoveUserTypeLibReferences(
&    System.int NCount,
&    System.String^% BstrTlbRef
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NCount

Number of user-specified type library references

BstrTlbRef

Array of strings of the user-specified type library references

Return Value

True if the user-specified type library references are removed, false if not

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IRemoveUserTypeLibReferences](#).

Remarks

Before calling this method, call [ISldWorks::GetUserTypeLibReferenceCount](#) to get the value for NCount.

Any macro recorded after this method is called will not have references to the removed user-specified type libraries.

An add-in can use this method to remove its type library references when the add-in is unloaded.

See Also

[ISldWorks Interface](#)

[ISldWorks Members](#)

[ISldWorks::IGetUserTypeLibReferences Method](#)

[ISldWorks::ISetUserTypeLibReferences Method](#)

[ISldWorks::RemoveUserTypeLibReferences Method](#)

[ISldWorks::UserTypeLibReferences Property](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IsBackgroundProcessingCompleted Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IsBackgroundProcessingCompleted Method (ISldWorks)

Gets whether SOLIDWORKS has finished background processing a drawing document that requires a lot of CPU time to open.

.NET Syntax

Visual Basic (Declaration)

```
Function IsBackgroundProcessingCompleted( _
    ByVal FilePath As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISldWorks
Dim FilePath As System.String
Dim value As System.Boolean
```

```
value = instance.IsBackgroundProcessingCompleted(FilePath)
```

C#

```
System.bool IsBackgroundProcessingCompleted(  
    System.string FilePath  
)
```

C++/CLI

```
System.bool IsBackgroundProcessingCompleted(  
& System.String^ FilePath  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FilePath

Fully qualified path and filename of the drawing document that was opened in a background process

Return Value

True if background processing is finished, false if not

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IsBackgroundProcessingCompleted](#).

Remarks

SOLIDWORKS recommends that you only use this method with [ISldWorks::EnableBackgroundProcessing](#) and [IDrawingDoc::BackgroundProcessingOption](#) when you can quickly open a drawing document via the user interface, but that same drawing takes significantly more time to open programmatically.

To more efficiently and programmatically open a drawing document that requires a lot of CPU time and no user input:

- Set `ISldWorks::EnableBackgroundProcessing` to true.
- Use [ISldWorks Event BackgroundProcessingStartNotify](#) to handle the background processing start event.
- Open the drawing document by calling either [ISldWorks::OpenDoc6](#) or [ISldWorks::OpenDoc7](#).
- Set `IDrawingDoc::BackgroundProcessingOption` to [swBackgroundProcessOption_e.swBackgroundProcessing_DeferToApplication](#).
- Call `ISldWorks::IsBackgroundProcessingCompleted` repeatedly, which polls the status of the open operation, to know when background processing ends.
- Use [ISldWorks Event BackgroundProcessingEndNotify](#) to handle the background processing end event.
- When the open operation is finished, set `ISldWorks::EnableBackgroundProcessing` to false.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsCommandEnabled Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : IsCommandEnabled Method (ISldWorks)

Gets whether the specified command is enabled.

.NET Syntax

Visual Basic (Declaration)

```
Function IsCommandEnabled( _  
    ByVal CommandID As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ISldWorks  
Dim CommandID As System.Integer  
Dim value As System.Boolean
```



```
value = instance.IsCommandEnabled(CommandID)
```

C#

```
System.bool IsCommandEnabled(  
    System.int CommandID  
)
```

C++/CLI

```
System.bool IsCommandEnabled(  
& System.int CommandID  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CommandID
Command ID as defined [swCommands_e](#) (see **Remarks**)

Return Value

True if the command is enabled, false if not

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::IsCommandEnabled](#).

Remarks

Before using this method, you must add a reference to the version-specific **SOLIDWORKS Commands type library** to access the SOLIDWORKS commands.

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::RunCommand Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetSelectionFilters Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : ISetSelectionFilters Method (ISldWorks)

Sets the status for multiple selection filters.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetSelectionFilters(  
    ByVal Count As System.Integer, _  
    ByRef SelType As System.Integer, _  
    ByVal State As System.Boolean _  
)
```

Visual Basic (Usage)

```
Dim instance As ISldWorks  
Dim Count As System.Integer  
Dim SelType As System.Integer  
Dim State As System.Boolean
```



```
instance.ISetSelectionFilters(Count, SelType, State)
```

C#

```
void ISetSelectionFilters(  
    System.int Count,  
    ref System.int SelType,  
    System.bool State  
)
```

C++/CLI

```
void ISetSelectionFilters(  
& System.int Count,  
& System.int% SelType,  
& System.bool State  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of values in SelType

SelType

- in-process, unmanaged C++: Pointer to an array of long values representing the [swSelectType_e](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

State

Visual Basic for Applications (VBA) Syntax

See [ISldWorks::ISetSelectionFilters](#).

Remarks

The state is applied to all elements in the SelType array. To determine if the selection filter is active, use [ISldWorks::GetApplySelectionFilter](#).

See Also

[ISldWorks Interface](#)
[ISldWorks Members](#)
[ISldWorks::GetSelectionFilter Method](#)
[ISldWorks::GetSelectionFilters Method](#)
[ISldWorks::IGetSelectionFilters Method](#)
[ISldWorks::IGetSelectionFiltersCount Method](#)
[ISldWorks::SetSelectionFilter Method](#)
[ISldWorks::SetSelectionFilters Method](#)
[ISldWorks::SetApplySelectionFilter Method](#)
[ISldWorks::IGetSelectionFiltersCount Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetUserTypeLibReferences Method (ISldWorks)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISldWorks Interface](#) : ISetUserTypeLibReferences Method (ISldWorks)

Sets the user-specified type library references.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetUserTypeLibReferences( _  
    ByVal NCount As System.Integer, _
```

```
ByRef BstrTlbRef As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As ISldWorks  
Dim NCount As System.Integer  
Dim BstrTlbRef As System.String  
  
instance.ISetUserTypeLibReferences(NCount, BstrTlbRef)
```

C#

```
void ISetUserTypeLibReferences(  
    System.int NCount,  
    ref System.string BstrTlbRef  
)
```

C++/CLI

```
void ISetUserTypeLibReferences(  
& System.int NCount,  
& System.String^% BstrTlbRef  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NCount

Number of user-specified type library references

BstrTlbRef

- in-process, unmanaged C++: Pointer to an array of strings of the user-specified type library references
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

- An add-in can use this method to add its type library references after the add-in is loaded.
- A user-specified type library first appears on the list of available references only after adding it and only after recording a macro.
- User-specified type library references are not persistent across SOLIDWORKS Design sessions.
- Only macros created after adding a user-specified type library reference can reference that type library.

See [Type Libraries](#) for details.

See Also

[ISldWorks Interface](#)

[ISldWorks Members](#)

[ISldWorks::GetUserTypeLibReferenceCount Method](#)

[ISldWorks::IGetUserTypeLibReferences Method](#)

[ISldWorks::IRemoveUserTypeLibReferences Method](#)

[ISldWorks::RemoveUserTypeLibReferences Method](#)

[ISldWorks::UserTypeLibReferences Property](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0